

A Study on Modern Compilers

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Compilers are complex software systems that translate high-level programming languages into machine code.

The first step in this process is lexical analysis. During lexical analysis, the compiler reads the source code

character by character and groups them into meaningful sequences called tokens.

This phase is critical because it simplifies the subsequent syntax analysis phase.

If there are any illegal characters, the lexical analyzer will flag them immediately and halt the compilation process.

Overall, the design of a lexical analyzer must be highly efficient, as it processes every single character of the input program.